Urban Heat Mitigation — AI/ML Resource Map

A comprehensive intelligence report for building a geospatial AI/ML pipeline to identify heat stress hotspots, quantify urban heating drivers, and optimize cooling interventions.

> Compiled June 2026 | Scoped to: LST mapping · driver attribution · physics-informed ML · scenario simulation · explainability · intervention ranking

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SECTION 1 — GITHUB REPOSITORIES (15)

R-01 · `ArhamOrioner/UHI-Analysis`

Type: Repository — Modular Python UHI analysis framework **URL:** https://github.com/ArhamOrioner/UHI-Analysis **What it does:** A full command-line-driven, city-portable UHI detection pipeline. Pulls MODIS/Landsat 8 LST + ESA WorldCover + Sentinel-2 NDVI/NDBI + SRTM DEM via GEE. Performs K-Means clustering of temperature zones, Getis-Ord Gi* hotspot analysis, Random Forest regression for LST prediction, and SHAP explainability for driver attribution. Applied to Las Vegas 2024. **Key variables:** LST, NDVI, NDBI, land cover, elevation, slope, aspect **Scenario simulation:** Hotspot detection only; no intervention simulation **Runnable:** Yes — Python CLI with `app.py`; `scikit-learn`, `geopandas`, `shap` stack **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best general-purpose starting repo. Has SHAP attribution built in.

R-02 · `yotkadata/uhi_detector`

Type: Repository — Deep Learning UHI detector with Streamlit app **URL:** https://github.com/yotkadata/uhi_detector **What it does:** Computes LST, NDVI, and emissivity from Landsat 8; uses a U-Net model to segment building footprints from satellite imagery (86% IoU on test set); combines with OSM building footprints; includes a live Streamlit demo for urban heat visualization. **Key variables:** LST (Landsat 8 B4/B5/B10), NDVI, emissivity, building footprints **Scenario simulation:** Hotspot detection only **Runnable:** Yes — Python 3.10, GDAL, Streamlit; pip installable **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Excellent for building footprint pipeline + LST + live demo.

R-03 · `Somnathab3/EYDS2025`

Type: Repository — EY 2025 Urban Heat Island Challenge winner pipeline **URL:** https://github.com/Somnathab3/EYDS2025 **What it does:** Integrated 260+ engineered features covering urban morphology, spectral indices, atmospheric conditions, socioeconomic characteristics. Landsat 8, Sentinel-2, Sentinel-3, ERA5, building data. Final model: ExtraTreesRegressor with $R^2 = 0.9822$ on validation. **Key variables:** L8_ST_B10 LST, NDVI, NDWI, Moisture Index, air temperature, relative humidity, wind speed, solar flux, Sentinel-3 brightness temperature, building density **Scenario simulation:** Hotspot prediction only **Runnable:** Yes — Jupyter notebooks with pip installs **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best feature engineering reference for the exact competition structure.

R-04 · `Kaamil008/2025-EY-Open-Science-AI-and-Data-Challenge-Cooling-Urban-Heat-Islands`

Type: Repository — XGBoost pipeline for UHI index prediction **URL:** https://github.com/Kaamil008/2025-EY-Open-Science-AI-and-Data-Challenge-Cooling-Urban-Heat-Islands **What it does:** Optimized XGBoost model trained on Sentinel-2 bands + Landsat LST + UTFVI (Urban Thermal Field Variance Index). Selects B01, B06, B11, NDVI, LST, UTFVI as key features. Compact, production-style notebook. **Key variables:** Sentinel-2 bands, NDVI, LST, UTFVI, building footprints **Scenario simulation:** Prediction only **Runnable:** Yes — notebook with pip install **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Compact, fast reference for XGBoost + UTFVI feature.

R-05 · `ErDipeshPaudel/EY-Open-Science-AI-Data-Challenge`

Type: Repository — Complete EY 2025 guide with Random Forest approach **URL:** https://github.com/ErDipeshPaudel/EY-Open-Science-AI-Data-Challenge **What it does:** Step-by-step guide to the 2025 EY UHI challenge. Planetary Computer raster sampling,

building density features via GeoPandas/Shapely, Random Forest Regressor from sklearn. Includes data loading from EY Training CSV (with updated dataset note). **Key variables:** UHI index, air temperature, relative humidity, wind, solar flux, Sentinel-2, building density **Scenario simulation:** None **Runnable:** Yes — Jupyter notebooks **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best beginner-oriented walkthrough; good for quick prototyping.

R-06 · `KonlavachMengsuwan/-Urban-Heat-Island-Mapping-with-Machine-Learning-using-Google-Earth-Engine`

Type: Repository — GEE Random Forest UHI mapping **URL:** https://github.com/KonlavachMengsuwan/-Urban-Heat-Island-Mapping-with-Machine-Learning-using-Google-Earth-Engine **What it does:** Full GEE JavaScript workflow: loads Landsat 8 Collection 2 Level-2 for summer months, computes LST, derives NDVI and NDBI, trains Random Forest to classify heat stress zones (cool → extreme). Highly customizable; works for any city with a bounding box change. **Key variables:** LST, NDVI, NDBI (all from Landsat 8) **Scenario simulation:** Heat stress classification (5 zones) **Runnable:** Yes — pure GEE JavaScript (no install) **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best zero-install entry point for city-scale LST + heat zone mapping.

R-07 · `wri/UrbanHeatMitigation`

Type: Repository — ML model for surface reflectivity estimation (cool roofs / streets) **URL:** https://github.com/wri/UrbanHeatMitigation **What it does:** World Resources Institute repo. Creates and applies ML models to estimate surface reflectivity of roofs and streets across US urban areas. Directly relevant to cool roof simulation and albedo-based intervention modeling. **Key variables:** Albedo, roof reflectivity, street reflectivity, satellite imagery **Scenario simulation:** Partial — estimates albedo baselines for intervention scenarios **Runnable:** Python 3.6; requires Descartes Labs API (registration needed) **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Unique albedo/reflectivity angle; best cool-roof data layer generator.

R-08 · `UMEP-dev/UMEP`

Type: Repository — Urban Multi-scale Environmental Predictor (QGIS plugin) **URL:** https://github.com/UMEP-dev/UMEP **What it does:** The most comprehensive open-source urban climate toolkit. Integrates SOLWEIG (mean radiant temperature), SUEWS (urban energy balance), TreePlanter (tree placement optimization), and >20 preprocessing tools. Accepts DSM, DEM, building height, canopy height, land cover, and meteorological inputs. Calculates UTCI, PET, Tmrt at 1m resolution. **Key variables:** DSM/DEM, building height, canopy height, land cover, air temperature, humidity, radiation **Scenario simulation:** YES — tree planting, green space, scenario comparison **Runnable:** Yes — QGIS plugin (Python)

License: GPL v3 **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Gold standard for physics-based scenario simulation. Best for SOLWEIG + TreePlanter.

R-09 · `UMEP-dev/solweig`

Type: Repository — SOLWEIG reimplemented in Rust (fast, GPU-optional) **URL:** https://github.com/UMEP-dev/solweig **What it does:** New Rust implementation of SOLWEIG (Solar and Longwave Environmental Irradiance Geometry). Computes mean radiant temperature (Tmrt), UTCI, PET at city scale. GPU-accelerated. Takes building height model + weather data → high-resolution maps of heat stress. With optional QGIS plugin. **Key variables:** Building height model, tree canopy, weather (air temp, humidity, radiation) **Scenario simulation:** YES — can compare street designs, tree planting, cool-roof strategies **Runnable:** Yes — Rust crate + QGIS plugin **License:** GPL v3 **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Fastest SOLWEIG; ideal for large-city scenario runs.

R-10 · `martibosch/lausanne-heat-islands`

Type: Repository — InVEST Urban Cooling Model application (full paper replication) **URL:** https://github.com/martibosch/lausanne-heat-islands **What it does:** Complete code for a published paper simulating urban heat islands in Lausanne using the InVEST Urban Cooling Model. Includes LULC preprocessing, evapotranspiration raster generation, biophysical table setup, urban greening scenario generation (tree canopy scenarios), calibration, and air temperature output. **Key variables:** LULC raster (10m), reference evapotranspiration, building density, tree canopy fraction **Scenario simulation:** YES — greening scenarios with tree canopy increase at different spatial distributions **Runnable:** Yes — Python + InVEST; conda installable **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best end-to-end InVEST scenario simulation with published validation.

R-11 · `martibosch/invest-ucm-calibration`

Type: Repository — Automated InVEST Urban Cooling Model calibration **URL:** https://github.com/martibosch/invest-ucm-calibration **What it does:** Python library for automated calibration of InVEST UCM using simulated annealing. Optimizes parameters (rmix, dcool, Wa, We, Ws) against measured station temperatures. Conda installable. **Key variables:** InVEST UCM parameters, station temperatures, LULC **Scenario simulation:** YES — supports before/after scenario comparison **Runnable:** Yes — pip/conda installable **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Essential if using InVEST for quantitative temperature reduction estimates.

R-12 · `palubad/LST-downscaling-to-10m-GEE`

Type: Repository — Landsat 8 LST downscaled to 10m using Sentinel-2 in GEE **URL:** https://github.com/palubad/LST-downscaling-to-10m-GEE **What it does:** GEE JavaScript + app that downscales Landsat 8 LST (30m) to 10m using Sentinel-2 NDVI, NDBI, NDWI via linear regression. $R^2 = 0.91$ between observed and predicted. Works globally for any city's bounding box. **Key variables:** Landsat 8 LST (30m) → Sentinel-2 spectral indices → 10m LST **Scenario simulation:** Hotspot detection only **Runnable:** Yes — GEE JavaScript app (zero install) **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Critical for creating the 10m LST input layer needed for all downstream models.

R-13 · `ibm-granite/granite-geospatial-land-surface-temperature`

Type: Repository + Pretrained Model — Foundation model for LST prediction **URL:** https://github.com/ibm-granite/granite-geospatial-land-surface-temperature **HuggingFace:** https://huggingface.co/ibm-granite/granite-geospatial-land-surface-temperature **What it does:** IBM Granite SWIN-Transformer fine-tuned on HLS L30 (Harmonized Landsat-Sentinel 30m) + ERA5-Land to predict LST at 30m spatial resolution and hourly temporal frequency. Trained on 28 global cities (2013–2023). Includes notebooks for getting started, data download, preprocessing, and temporal gap-filling ("tweening"). **Key variables:** HLS L30 imagery, ERA5-Land 2m temperature → 30m hourly LST **Scenario simulation:** LST prediction only; no intervention **Runnable:** Yes — terratorch + HuggingFace hub; pip installable **License:** Apache 2.0 (IBM open source) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best pretrained LST model. Use for filling LST time gaps and improving temporal coverage.

R-14 · `kunifujiwara/VoxCity`

Type: Repository — 3D city model generator + urban environment simulator **URL:** https://github.com/kunifujiwara/VoxCity **What it does:** Python package that automatically downloads building heights, tree canopy heights, land cover, and terrain elevation for any city worldwide and generates an integrated voxel city model. Simulator subpackage runs solar irradiance, sky view index, and green view index. Exports to ENVI-met (INX), Blender, Rhino (OBJ) for microclimate simulation. **Key variables:** Building heights, canopy heights, land cover, terrain; solar radiation, view factors **Scenario simulation:** YES — exports to ENVI-met for full microclimate scenario runs **Runnable:** Yes — pip installable Python package **License:** MIT **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best automated city model prep pipeline; connects directly to SOLWEIG/ENVI-met.

R-15 · `JZembower/Urban_Heat_Island_Index_Prediction_Model`

Type: Repository — EY 2025 Challenge ($R^2 = 0.9606$, ranked 86th globally) **URL:** https://github.com/JZembower/Urban_Heat_Island_Index_Prediction_Model **What it does:** ML pipeline for NYC UHI index prediction. Used satellite imagery, geospatial data, feature

importance maps, and heatmap visualization. Documents the full analytical process including Final_Model, NYC-specific notebook, and Greensburg transferability notebook. Demonstrates model generalizability across locations. **Key variables:** Satellite imagery-derived features, UHI index **Scenario simulation:** Prediction + transferability analysis **Runnable:** Yes — Jupyter notebooks **License:** Not specified **Suitability:** ⭐ ⭐ ⭐ MEDIUM-HIGH — Good for generalizability testing and cross-city validation.

SECTION 2 — OPEN DATASETS (10)

D-01 · Landsat 8 Collection 2 Level-2 — Thermal Band (Band 10 / ST_B10)

Type: Satellite raster dataset **Access:** Google Earth Engine (`LANDSAT/LC08/C02/T1_L2`), USGS EarthExplorer (earthexplorer.usgs.gov) **Why it matters:** Primary source of 30m Land Surface Temperature. Band ST_B10 with formula `ST_B10 × 0.00341802 + 149.0 − 273.15` gives LST in °C. Global, free, 16-day revisit since 2013. **Variables included:** LST, surface reflectance (B1–B7), NDVI-derivable, cloud mask **Scenario support:** Foundation layer for all hotspot mapping **License:** Public domain (US government data) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH

D-02 · ECOSTRESS Land Surface Temperature & Emissivity (ECO2LSTE)

Type: Satellite dataset **Access:** NASA Earthdata (earthdata.nasa.gov/sensors/ecostress), GEE community catalog **Why it matters:** 70m LST at 1–5 day revisit with overpass during afternoon peak heat (key for heat stress mapping). Much higher temporal resolution than Landsat. Emissivity data included. **Variables included:** LST (70m), emissivity, quality flags **Scenario support:** Hotspot detection, temporal monitoring **License:** Public domain (NASA) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Particularly valuable for peak-heat temperature analysis.

D-03 · ERA5 / ERA5-Land Reanalysis (Copernicus CDS)

Type: Meteorological reanalysis dataset **Access:** https://cds.climate.copernicus.eu | Python API: `pip install cdsapi` **Why it matters:** Hourly air temperature (2m), relative humidity, wind speed/direction, solar radiation, and >200 variables at 0.25° (~28km) resolution since 1940. The standard atmospheric driver dataset for urban heat models. Used in IBM Granite LST model and most UHI ML pipelines. **Variables included:** 2m air temperature (t2m), 2m dewpoint (d2m), 10m u/v wind, surface solar radiation downwards, skin temperature, total precipitation **Scenario support:** Atmospheric boundary conditions for scenario simulations **License:** Copernicus License (free for research/commercial use with attribution) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH

D-04 · GHSL Global Human Settlement Layer 2023 (GHS-BUILT-H, GHS-BUILT-S, GHS-SMOD)

Type: Built environment morphology dataset **Access:** https://human-settlement.emergency.copernicus.eu/datasets.php | JRC EU Data Catalogue **Why it matters:** Global building heights (100m and 10m), built-up surface, volume, population, and settlement classification for 1975–2020 in 5-year intervals. Also includes GHS-OBAT (Open Building Attribute Table) with individual building heights, compactness, function, construction year linked to Overture Maps footprints. Available as GEE community dataset. **Variables included:** Average net building height (ANBH), built-up surface fraction, built-up volume, population density, settlement type (urban/suburban/rural) **Scenario support:** Urban morphology layer; baseline for intervention scenarios **License:** CC BY 4.0 **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Essential for building density, height, and urban form features.

D-05 · ESA WorldCover 10m Land Cover (2020, 2021)

Type: Land cover classification raster **Access:** GEE (`ESA/WorldCover/v200`), ESA WorldCover portal, STAC **Why it matters:** 10m global land cover with 11 classes (trees, shrubland, grassland, cropland, built-up, bare/sparse, snow, water, wetland, mangroves, moss). Critical for LULC classification and impervious surface mapping. Better than Landsat-derived LULC for urban use. **Variables included:** Land cover class (11 classes), tree/built-up/impervious masks **Scenario support:** Input to InVEST UCM, SOLWEIG, and ML feature engineering **License:** CC BY 4.0 **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH

D-06 · OpenStreetMap (OSM) Urban Features

Type: Vector urban infrastructure dataset **Access:** `osmnx` Python library, Overpass API (overpass-api.de), GeoFabrik downloads (geofabrik.de) **Why it matters:** Building footprints, road networks, green spaces, water bodies, land use polygons. Free, global, community-maintained. Essential for urban morphology features (building density, road density, green space fraction). **Variables included:** Building footprints, roads, parks, water bodies, land use areas **Scenario support:** Baseline layer; can modify LULC to simulate greening **License:** Open Database License (ODbL) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best source for building footprints + road + green space layers.

D-07 · Kaggle — UHI Monitoring Dataset (Atharva Soundankar)

Type: Tabular dataset **URL:** https://www.kaggle.com/datasets/atharvasoundankar/urban-heat-island-uhi-monitoring-dataset **What it contains:** Urban monitoring data with UHI Index values, atmospheric variables, and spatial features suitable for exploratory data

analysis and ML modeling. Includes accompanying analysis notebook. **Variables included:** UHI Index, air temperature, humidity, wind, spectral indices **Scenario support:** Prediction modeling baseline **License:** CC0 / Public Domain **Suitability:** ⭐ ⭐ ⭐ MEDIUM — Good for quick EDA and baseline models.

D-08 · Kaggle — Global Urban Heat Island Dataset

Type: Multi-city tabular/spatial dataset **URL:** https://www.kaggle.com/datasets/riccardomogavero/global-urban-heat-island-uhi-data-set **What it contains:** Global UHI intensity data across multiple cities with urban and rural temperature comparisons. **Variables included:** UHI intensity, city metadata **Scenario support:** Comparative analysis **License:** CC BY 4.0 **Suitability:** ⭐ ⭐ ⭐ MEDIUM — Useful for cross-city validation.

D-09 · EY Open Science Challenge 2025 — Training Data (UHI Index with NYC Ground Truth)

Type: Challenge dataset — labeled UHI ground truth **Access:** https://challenge.ey.com/challenges/the-2025-ey-open-science-ai-and-data-challenge-cooling-urban-heat-islands-external-participants/data-description **What it contains:** Ground-truth UHI Index labels with geographic coordinates for New York City + Sentinel-2 + Sentinel-3 + Landsat 8 imagery layers + 3D building data. The richest labeled dataset for UHI regression. **Variables included:** UHI index (target), Sentinel-2 bands + indices, Sentinel-3 brightness temperatures, Landsat 8 LST, building footprints, air temperature, humidity, wind, solar flux **Scenario support:** Prediction (labeled training data) **License:** Challenge terms (non-commercial) **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best labeled dataset with satellite + met + building features integrated.

D-10 · NASA/USGS HLS L30 (Harmonized Landsat-Sentinel 2, 30m)

Type: Harmonized multispectral satellite dataset **Access:** NASA LP DAAC (lpdaac.usgs.gov/products/hlsl30v002/), Microsoft Planetary Computer **Why it matters:** Harmonized Landsat 8 and Sentinel-2 reflectance at 30m resolution, unified band naming, 2–3 day revisit. Basis of IBM Granite LST model. Superior to either sensor alone for dense time series. **Variables included:** Surface reflectance (all bands), cloud mask, view angles **Scenario support:** Feature input for time-series LST models and ML training **License:** Public domain (US government) **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Key dataset for the IBM Granite LST pipeline.

SECTION 3 — NOTEBOOKS & DEMOS (10)

N-01 · EY 2025 Challenge — Getting Started Notebook (Planetary Computer)

URL: https://challenge.ey.com (linked from data description page) **What it shows:** How to access and combine the EY challenge data — Sentinel-2 rasters via STAC API, Landsat 8 LST band, building footprints — in Python using `rasterio`, `rioxarray`, `pystac-client`. Official baseline. **Runnable:** Yes — Jupyter; requires Planetary Computer or local install **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Start here for the competition pipeline.

N-02 · IBM Granite LST — Getting Started Notebook

URL: https://github.com/ibm-granite/granite-geospatial-land-surface-temperature/blob/main/notebooks/1_getting_started.ipynb **What it shows:** How to load the pretrained IBM Granite SWIN-LST model from HuggingFace, prepare HLS + ERA5 inputs, run inference for LST prediction at 30m. **Runnable:** Yes — `pip install -e .` in repo **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best foundation model notebook; gives you hourly 30m LST for any city.

N-03 · IBM Granite LST — Tweening (Temporal Gap-Filling) Notebook

URL: https://github.com/ibm-granite/granite-geospatial-land-surface-temperature/blob/main/notebooks/4_introduction_to_LST_Tweening.ipynb **What it shows:** Synthesizing hourly LST between satellite overpasses using ERA5 temperature statistics as drivers. Solves the temporal limitation of Landsat (16-day revisit). **Runnable:** Yes **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Critical for creating dense LST time series without cloud gaps.

N-04 · Kaggle — Urban Heat Island Analysis & Prediction Notebook

URL: https://www.kaggle.com/code/devraai/urban-heat-island-analysis-prediction **What it shows:** EDA and prediction modeling on the UHI Monitoring Dataset. Includes feature correlation, scatter plots, and ML model comparison. **Runnable:** Yes — Kaggle kernel (no install) **Suitability:** ⭐ ⭐ ⭐ MEDIUM — Good starting point for data exploration.

N-05 · GEE Landsat 8 LST Computation Scripts (Parastatidis / KIT)

URL: https://code.earthengine.google.com/collection/projects/sat-imagery/lst-algorithms (referenced from MDPI 2020 paper) **What it shows:** Full SMW algorithm for LST retrieval from Landsat 4/5/7/8 using NCEP TCWV, ASTER emissivity, NDVI-based FVC correction. GEE JavaScript scripts. **Runnable:** Yes — GEE JavaScript; no install **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Most physically rigorous open LST algorithm in GEE; citable.

N-06 · LST Downscaling GEE App (palubad)

URL: https://github.com/palubad/LST-downscaling-to-10m-GEE (GEE app linked from repo) **What it shows:** Interactive GEE web app for downscaling Landsat 8 LST to 10m for any city. Live demo mode; can be forked and deployed. **Runnable:** Yes — GEE JavaScript app **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Instant 10m LST for any city; zero friction.

N-07 · UMEP — Introduction to SOLWEIG Tutorial

URL: https://umep-docs.readthedocs.io/projects/tutorial/en/latest/Tutorials/IntroductionToSolweig.html **What it shows:** Step-by-step tutorial running SOLWEIG in QGIS: loading DSM, CDSM, DEM, land cover raster; configuring meteorological inputs; computing Tmrt and UTCI maps; visualizing results. Uses Gothenburg sample data. **Runnable:** Yes — QGIS + UMEP plugin **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best onboarding for SOLWEIG physics-based simulation.

N-08 · UMEP — TreePlanter Tutorial (Tree Placement Optimization)

URL: https://github.com/UMEP-dev/UMEP-Tutorials/blob/master/docs/source/Tutorials/IntroductionToTreePlanter.rst **What it shows:** How to run UMEP TreePlanter after a SOLWEIG simulation to find optimal tree placement locations that minimize mean radiant temperature within a designated planting area. **Runnable:** Yes — QGIS + UMEP plugin **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Directly answers "where should trees go to reduce heat the most?"

N-09 · ERA5 Python Download + Visualization Notebooks (loicduffar)

URL: https://github.com/loicduffar/ERA5-tools **What it shows:** Jupyter notebooks for ERA5-Land download, NetCDF visualization, time-series extraction at specific pixels, map animation, and monthly/hourly analysis. **Runnable:** Yes — `pip install cdsapi xarray` **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Fastest way to set up ERA5 pipeline for meteorological driver data.

N-10 · VoxCity 3D City Model Demo (kunifujiwara)

URL: https://github.com/kunifujiwara/VoxCity (notebooks in `/examples/`) **What it shows:** Automated generation of 3D voxel city models for any city; solar irradiance computation; sky view index; export to ENVI-met for microclimate simulation. Demonstrates the workflow for 8 global cities. **Runnable:** Yes — `pip install voxcity` **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Key preprocessing step before SOLWEIG/ENVI-met scenario runs.

SECTION 4 — PAPERS WITH CODE / BENCHMARK IMPLEMENTATIONS (5)

P-01 · PINN Framework for Urban Heat + Green Infrastructure Quantification

Title: Smart Urban Design with Physics-Informed Neural Networks: Quantifying Temperature Reductions from Green Infrastructure Using Satellite Thermal Data **URL:** https://www.researchgate.net/publication/396689873 **What it implements:** A PINN that embeds heat diffusion equations directly into the loss function alongside satellite-derived LST, NDVI, and emissivity, plus vector building/road data in GeoJSON. Applied to Bologna and Washington DC. Green roof + tree interventions reduced hotspot extent by 97–99% with average temp reductions of 1.3–1.8°C and max local cooling of 6–8°C. **Code status:** Paper includes full methodology; code link expected in supplementary **Benchmark:** R^2 = 0.926–0.937 vs. Landsat 8 validation **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Direct implementation reference for the physics-informed ML + intervention component.

P-02 · GSM-UTCI Multimodal Deep Learning for UTCI Prediction + Greening Scenarios

Title: GSM-UTCI: A Multimodal Deep Learning Framework for High-Resolution Urban Thermal Comfort Prediction and Greening Scenario Analysis in Philadelphia **URL:** https://arxiv.org/abs/2507.23000 **What it implements:** SOLWEIG for Tmrt + deep learning for UTCI prediction from street view + satellite + land cover. Scenario analysis comparing tree canopy, impervious surface conversion, and other interventions. **Code status:** Preprint (July 2025); code likely in supplementary **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best recent reference for UTCI-based scenario analysis with ML.

P-03 · InVEST Urban Cooling Model Calibration + Paris Greening Scenario (Code)

Title: Calibrating and validating the InVEST urban cooling model: Case studies in France and the US **URL:** https://gmd.copernicus.org/articles/17/4755/2024/ **Code:** https://github.com/martibosch/invest-ucm-calibration/tree/v0.6.0 | Zenodo archive linked in paper **What it implements:** Full calibration algorithm for InVEST UCM. Paris metropolitan greening scenario (50% surface coverage with grass + trees) simulated and validated. r^2 = 0.55/0.85 for day/night temperature vs. physics-based alternative. **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Best validated InVEST scenario + code; directly citable for temperature reduction claims.

P-04 · XGBoost-SHAP Urban Heat Drivers across Local Climate Zones

Title: How do built environment factors influence urban heat islands across local climate zones? Evidence from an interpretable XGBoost-SHAP model **URL:** https://www.sciencedirect.com/science/article/abs/pii/S2210670725009485 (Sustainable Cities and Society, 2025) **What it implements:** XGBoost + SHAP for attributing LST to urban form factors (building height, coverage ratio, sky view factor, NDVI, albedo, NLI) across LCZ types. SHAP interaction values reveal which drivers dominate in dense vs. open urban areas. **Code status:** Published paper; implementation follows standard sklearn + shap patterns **Benchmark:** R^2 = 0.80–0.95 on training/test **Suitability:** ⭐ ⭐ ⭐ ⭐ ⭐ HIGH — Key reference for the explainability + driver attribution component.

P-05 · Disentangling Climatic and Surface Drivers with XAI (GEE + SHAP across 8 cities)

Title: Disentangling Climatic and Surface-Physical Drivers of the Urban Heat Island Using Explainable AI Across U.S. Cities **URL:** https://doi.org/10.3390/su18083694 (MDPI Sustainability, April 2026) **What it implements:** Standardized XAI framework in GEE analyzing census-tract LST and SUHII across 8 climatically diverse US cities (2018–2024). XGBoost best performer (SUHII R^2 = 0.879, RMSE not given). City-held-out cross-validation + SHAP for cross-city driver comparison. **Code status:** GEE-based; paper available; replication follows standard GEE + XGBoost workflow **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best cross-city driver attribution framework; directly applicable to Indian city contexts.

SECTION 5 — SIMULATION TOOLS & FRAMEWORKS (5)

T-01 · UMEP / SOLWEIG + TreePlanter (Urban Multi-scale Environmental Predictor)

URL: https://umep-docs.readthedocs.io | https://github.com/UMEP-dev/UMEP **What it does:** Full urban microclimate physics suite. SOLWEIG computes mean radiant temperature from 3D urban geometry + weather. TreePlanter optimizes tree placement to minimize Tmrt. SUEWS simulates urban energy balance. All operate at 1m resolution. **Inputs:** DSM, DEM, building height, canopy height model, land cover, hourly meteorology **Outputs:** Tmrt maps, UTCI maps, shading maps, heat stress risk maps, tree placement recommendations **Scenario support:** YES — tree cover, vegetation, surface albedo scenarios **Use in pipeline:** Stage 3 (physics-based calibration), Stage 4 (simulation), Stage 5 (scenario evaluation)

T-02 · InVEST Urban Cooling Model (Natural Capital Project)

URL: https://naturalcapitalproject.stanford.edu/software/invest-models/urban-cooling **Documentation:** https://invest.readthedocs.io **What it does:** Estimates air temperature reduction from vegetation based on shade, evapotranspiration, albedo, and distance from cooling islands. Computes heat mitigation index, wet bulb globe temperature, energy savings, and work productivity loss. Calibratable to local temperature station data. **Inputs:** LULC raster, reference evapotranspiration, biophysical table (shade/Kc/albedo per LULC class), optional building footprints **Outputs:** Heat mitigation index raster, air temperature map (°C), energy savings, WBGT **Scenario support:** YES — generate alternate LULC maps with greening, compute ΔT **Use in pipeline:** Stage 4 (intervention simulation), Stage 5 (scenario comparison)

T-03 · VoxCity (3D City Model Generator + Simulator)

URL: https://github.com/kunifujiwara/VoxCity **What it does:** Python package that builds 3D city models by automatically downloading building heights, tree canopies, land cover, and terrain; voxelizes them; and runs solar radiation + view index simulations. Exports to ENVI-met for microclimate simulation. **Inputs:** City name or bounding box → auto-downloads from global open data sources **Outputs:** Voxel city model, solar irradiance raster, sky view factor, green view index; INX file for ENVI-met **Scenario support:** Via ENVI-met export **Use in pipeline:** Stage 2 preprocessing (3D urban form), Stage 4 (simulation input prep)

T-04 · GeoPandas + Shapely + OSMnx Urban Form Feature Engineering

URL: https://github.com/gboeing/osmnx | https://geopandas.org **What it does:** `osmnx` fetches building footprints, street networks, and land use from OpenStreetMap for any city. `geopandas` + `shapely` compute morphology features: building density, floor area ratio, sky view factor proxy, road density, distance to green space, distance to water. **Use in pipeline:** Stage 2 (driver feature engineering from urban morphology) **Suitability:** ⭐⭐⭐⭐⭐ HIGH — Essential toolchain for extracting urban form features from OSM.

T-05 · SHAP (SHapley Additive exPlanations) Python Library

URL: https://github.com/slundberg/shap | `pip install shap` **What it does:** Model-agnostic explainability framework. For any trained ML model (RF, XGBoost, neural network), computes Shapley values showing the contribution of each feature to each prediction. Key outputs: feature importance plot, summary plot, dependence plot, interaction plot. **Use in pipeline:** Stage 3 (driver attribution, explainability), Stage 6 (intervention ranking) **Suitability:** ⭐⭐⭐⭐⭐ HIGH — Industry standard for the explainability requirement.

A-01 · WRI Urban Heat Mitigation — Surface Reflectivity Model Archive

URL: https://github.com/wri/UrbanHeatMitigation **Why it matters:** Legacy WRI repository (~2018–2020) with a complete workflow for estimating roof and street albedo across US cities using ML. Uses Descartes Labs API (now restricted). Contains the conceptual framework for how ML can identify cool roof intervention targets. **Status:** Partially runnable (Descartes Labs access limited); architecture fully reusable. **Suitability:** ⭐ ⭐ ⭐ MEDIUM — Use for cool roof modeling concepts; replace Descartes Labs with Planetary Computer.

A-02 · Lausanne Heat Island Preprint (bioRxiv 2020) + GitHub

URL: https://www.biorxiv.org/content/10.1101/2020.11.09.373779 | Code: https://github.com/martibosch/lausanne-heat-islands **Why it matters:** The seminal paper for InVEST urban cooling model + greening scenario simulation. Published in Royal Society Open Science. Includes LULC maps at 10m, evapotranspiration, biophysical tables, calibrated model, and greening scenarios (random, clustered, along parks). **Status:** Fully archived and reproducible **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Best citable precedent for InVEST-based intervention evaluation.

A-03 · UMEP-dev GitHub — All Tutorial Data + Goteborg Reference Dataset

URL: https://github.com/UMEP-dev/UMEP-Tutorials (data downloadable from docs) **Why it matters:** Official sample datasets for UMEP including DSM, DEM, CDSM, land cover rasters for Gothenburg. Used in all official tutorials for SOLWEIG, SUEWS, TreePlanter. Pre-validated inputs for testing the physics simulation pipeline. **Status:** Actively maintained **Suitability:** ⭐ ⭐ ⭐ ⭐ HIGH — Essential test data for validating UMEP/SOLWEIG setup before applying to target city.

SECTION 7 — PRACTICAL SYNTHESIS

7.1 Best Datasets for a City-Scale Pilot (Indian City Context)

LST layer: Landsat 8 C2L2 via GEE (30m, free, global) + LST downscaling to 10m via R-12 (palubad GEE app)

Land cover: ESA WorldCover 10m (GEE) — best free global LULC at 10m resolution

Urban morphology: GHSL 2023 building heights + OSM footprints via `osmnx` + road network density

Meteorological drivers: ERA5-Land via Copernicus CDS (2m temperature, humidity, wind, radiation)

Population/vulnerability: GHSL population raster (GHS-POP) for exposure analysis

Vegetation: NDVI time series from Landsat 8 or Sentinel-2 (GEE)

Water bodies: NDWI from Landsat/Sentinel-2; JRC Global Surface Water (GEE)

For competition-ready labeled data: EY 2025 challenge dataset (D-09)

7.2 Best Model Families by Task

Task	Recommended Model	Repo/Tool
LST hotspot detection	Random Forest / XGBoost + NDVI + NDBI + NDWI	R-06 (GEE), R-03
LST prediction (regression)	XGBoost or ExtraTrees with 100+ features	R-03, R-04
High-res LST from coarse	IBM Granite SWIN-Transformer	R-13
Driver attribution	XGBoost + SHAP interaction values	T-05, P-04
Physics-constrained LST	PINN with heat diffusion loss	P-01
Intervention scenario ΔT	InVEST UCM (fast) or SOLWEIG (precise)	T-01, T-02
Tree placement optimization	UMEP TreePlanter	T-01, N-08
Temporal LST gap filling	IBM Granite Tweening	N-03
Intervention explainability	SHAP waterfall/dependence plots	T-05

7.3 Tools Closest to Intervention Simulation

Tier 1 — Quantitative ΔT output:

- InVEST Urban Cooling Model (T-02) → outputs air temperature reduction in °C per greening scenario
- UMEP SOLWEIG + TreePlanter (T-01) → outputs Tmrt / UTCI reduction per tree placement scenario

Tier 2 — Spatial scenario comparison:

- Lausanne greening repo (R-10) → generates LULC alternate maps + simulates temperature distribution per scenario
- InVEST UCM calibration (R-11) → calibrates model to station data for quantitative ΔT validation

Tier 3 — PINN-based intervention:

- PINN framework (P-01) → modifies NDVI/emissivity in input to simulate green roof / tree planting → predicts new LST

Key insight: InVEST UCM is the fastest path to "estimated temperature reduction (°C)" per intervention type. SOLWEIG provides finer-grained pedestrian-level thermal comfort outputs. Combine both for the strongest submission.

7.4 Resource Map: 30-Hour Prototype vs. Full Research Build

30-HOUR PROTOTYPE PIPELINE

Hour 0–3: Data acquisition

- Pull Landsat 8 LST for target city via the GEE script from R-06 (copy-paste, works in 30 min)
- Pull ESA WorldCover LULC via GEE
- Download ERA5-Land air temperature + humidity for same dates via N-09 notebook

Hour 3–8: Feature engineering

- Compute NDVI, NDBI, NDWI, UTFVI in GEE
- Use `osmnx` to compute building density, road density, distance to green space
- Stack all rasters with `rioxarray` or `rasterio` into a feature matrix

Hour 8–16: Model training

- Train XGBoost or Random Forest on feature matrix → LST prediction
- Run SHAP to get feature importance and driver attribution plots
- Identify top 3–5 LST hotspot clusters using Getis-Ord Gi* (R-01 has this)

Hour 16–22: Scenario simulation

- Set up InVEST UCM (install with pip, tutorial at T-02)
- Run baseline scenario with current LULC
- Generate 2–3 greening scenarios (increase tree canopy in hotspots)
- Compute ΔT per scenario → ranked intervention table

Hour 22–30: Output generation

- LST hotspot map + heat stress zones (matplotlib / folium)
- SHAP summary plot (driver attribution)
- Scenario comparison table (intervention type, location, estimated °C reduction)
- Export GeoTIFF outputs for map layers

FULL RESEARCH BUILD (2–4 weeks)

Phase 1 — Data pipeline (Days 1–3)

- IBM Granite LST model (R-13) for hourly 30m LST with temporal gap-filling
- VoxCity (R-14) for automated 3D city model → SOLWEIG-ready inputs
- GHSL building heights + OSM morphology features (D-04 + T-04)
- ECOSTRESS for afternoon peak heat LST (D-02)

Phase 2 — Driver modeling (Days 4–7)

- XGBoost + 3D urban form features (building height variability, sky view factor, aspect ratio)
- SHAP interaction values for non-linear driver relationships
- Local Climate Zone (LCZ) stratification for zone-specific analysis
- Spatial lag features (e.g., distance to parks, density within 500m radius)

Phase 3 — Physics-informed ML (Days 8–12)

- PINN with surface energy balance constraints (P-01 methodology)
- InVEST UCM calibration against local weather station data (R-11)
- SOLWEIG Tmrt + UTCI mapping at 1m resolution for hotspot neighborhoods (T-01)

Phase 4 — Scenario simulation (Days 13–18)

- Generate greening LULC maps: 20%, 40%, 60% tree cover increase in hotspot zones
- Cool roof scenario: increase albedo of built-up class in InVEST biophysical table
- Water body scenario: add reflective water index to InVEST
- Run all scenarios through InVEST UCM → ΔT per scenario per pixel
- Run SOLWEIG for top 2–3 candidate hotspot neighborhoods with tree planting

Phase 5 — Optimization + output (Days 19–24)

- Spatial optimization: rank intervention locations by ΔT per unit area
- Cost-effectiveness scoring: ΔT / intervention complexity
- Generate output layers: hotspot map, driver attribution map, scenario ΔT maps

- Explainability report: SHAP plots, partial dependence plots, LCZ-stratified analysis
- Final recommendation: intervention type × location × estimated temperature reduction table

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SECTION 8 — KEY TECHNICAL NOTES

LST Computation (Landsat 8, GEE):

```javascript
var lst = landsat.select('ST_B10')
  .multiply(0.00341802)
  .add(149.0)
  .subtract(273.15); // Result in °C
```

NDVI:

```javascript
var ndvi = landsat.normalizedDifference(['SR_B5', 'SR_B4']);
```

NDBI (Built-up index):

```javascript
var ndbi = landsat.normalizedDifference(['SR_B6', 'SR_B5']);
```

UTFVI (Urban Thermal Field Variance Index):

```python
utfvi = (lst - lst_mean) / lst_mean
```

InVEST UCM Python call:

```python
```

```
from natcap.invest import urban_cooling_model
urban_cooling_model.execute({
    'lulc_raster_path': 'lulc.tif',
    'ref_eto_raster_path': 'eto.tif',
    'aoi_vector_path': 'city.gpkg',
    'biophysical_table_path': 'biophysical_table.csv',
    'cc_method': 'factors',
    'uhi_max': 3.5,  # calibrate to local data
    'workspace_dir': 'invest_output'
})
```

ERA5 Download (Python CDS API):

```
python

import cdsapi
c = cdsapi.Client()
c.retrieve('reanalysis-era5-land', {
    'variable': ['2m_temperature', '2m_dewpoint_temperature',
                 '10m_u_component_of_wind', '10m_v_component_of_wind',
                 'surface_solar_radiation_downwards'],
    'year': '2023', 'month': '06', 'day': ['01'],
    'time': [f'{h:02d}:00' for h in range(24)],
    'area': [city_N, city_W, city_S, city_E],
    'format': 'netcdf'
}, 'era5_output.nc')
```

APPENDIX — QUICK-START PRIORITY LIST

For a competition team starting today, in strict order of highest-to-lowest time-ROI:

1. **Open the EY 2025 challenge baseline** (N-01) → understand the data structure
2. **Run R-06 GEE script** for your city → instant LST + heat stress zone map (30 min)
3. **Clone R-03 (Somnathab3/EYDS2025)** → study the 260-feature engineering approach
4. **Set up N-09 (ERA5 tools)** → add atmospheric drivers (humidity, wind, temp)
5. **Add SHAP to your model** (T-05) → driver attribution output ready
6. **Install InVEST and run T-02** → first ΔT estimates from greening scenarios
7. **Run SOLWEIG via UMEP** (T-01, N-07) → physics-validated thermal comfort maps
8. **Use IBM Granite (R-13)** → fill LST temporal gaps; hourly predictions
9. **Build PINN on top** (P-01 methodology) → physics-informed intervention modeling
10. **VoxCity + ENVI-met** (R-14) → neighborhood-scale scenario visualization

Resource map compiled from web searches across GitHub, Kaggle, HuggingFace, Copernicus, arXiv, NASA, JRC/EU, NatCap, and UMEP documentation. All URLs verified June 2026.